COURSEDUP

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Linear Algebra I

Quadratic Forms, Principal Axes Theorem

INSTRUCTOR	UNIVERSITY	CODE
Sarah Collins	Virtual Earth University	GLL101
FACULTY	**DEPARTMENT**	**WEEK**
Science and Engineering	Mathematics	13
DISCIPLINE	**TYPE**	**LEVEL**
Mathematics	Quizzes	Undergraduate

THEORETICAL FRAMEWORK

- Quadratic Forms
- Symmetric Matrices
- Principal Axes Theorem
- Change of Basis
- Orthogonal Diagonalization
- Conic Sections and Quadric Surfaces

Quadratic Forms, Principal Axes Theorem

Learning Objectives:

- Represent quadratic forms using symmetric matrices.
- Apply the Principal Axes Theorem to perform orthogonal diagonalization.
- Classify conic sections and quadric surfaces based on canonical forms.
- Analyze the impact of change of basis on the representation of quadratic forms.

Duration: 45 minutes **Total:** 100 points

PART A: MULTIPLE CHOICE (50 POINTS — 5 PTS EACH)

Choose the single best answer for each question.

1. Given the quadratic form $Q(x_1, x_2) = 3x_1^2 + 4x_1x_2 + 6x_2^2$, what is the associated symmetric matrix A such that $Q(x) = x^T Ax$?

A. $[3, 4; 4, 6]$

B. $[3, 2; 2, 6]$

C. $[6, 4; 4, 12]$

D. $[3, 4; 0, 6]$

2. If a quadratic form $Q(x) = x^T Ax$ is subjected to an orthogonal change of variables $x = Py$, where P is an orthogonal matrix, what happens to the eigenvalues of A?

A. They are multiplied by $det(P)$.

B. They are transformed by $P^T AP$.

C. They remain unchanged.

D. They become the squares of the original eigenvalues.

3. A quadratic form is classified as positive definite if:

A. All eigenvalues of its symmetric matrix are positive.

B. The determinant of its symmetric matrix is negative.

C. At least one eigenvalue is zero.

D. The trace of its symmetric matrix is negative.

4. Consider the conic section defined by $5x^2 + 4xy + 2y^2 = 1$. The eigenvalues of the associated matrix are $\lambda_1 = 6$ and $\lambda_2 = 1$. What is the canonical form?

A. $6y_1^2 + y_2^2 = 1$

B. $x^2/6 + y^2/1 = 1$

C. $y_1^2/6 + y_2^2 = 1$

D. $6x^2 + 1y^2 = 1$

5. Which of the following is true regarding the Principal Axes Theorem?

A. It only applies to 2×2 matrices.

B. It states that every symmetric matrix is orthogonally diagonalizable.

C. It requires the matrix to be non-singular.

D. It ensures the resulting diagonal matrix has only ones on the diagonal.

6. The signature of a quadratic form is defined by:

A. The sum of the eigenvalues.

B. The number of positive, negative, and zero eigenvalues.

C. The product of the diagonal elements.

D. The trace of the matrix squared.

7. If a symmetric matrix A has a zero eigenvalue, the quadratic form is said to be:

A. Positive definite.

B. Negative definite.

C. Positive semi-definite or negative semi-definite (depending on other eigenvalues).

D. Indefinite.

8. What does the term 'orthogonal diagonalization' achieve for a quadratic form?

A. It eliminates the cross-product terms.

B. It rotates the axes to align with the eigenvectors.

C. It simplifies the expression to a sum of squares.

D. All of the above.

9. Given $Q(x) = x_1^2 - 4x_1x_2 + x_2^2$, the eigenvalues are 3 and -1. What is the geometric nature of the curve $Q(x) = 1$?

A. Ellipse.

B. Hyperbola.

C. Parabola.

D. Circle.

10. What is the primary purpose of the Principal Axes Theorem in the context of quadric surfaces?

A. To calculate the volume of the surface.

B. To identify the orientation and shape of the surface by removing cross-terms.

C. To determine the center of the surface.

D. To prove the surface is bounded.

PART B: TRUE / FALSE (20 POINTS — 4 PTS EACH)

Circle T if the statement is True or F if the statement is False.

1. [T / F] Every quadratic form can be associated with a unique symmetric matrix.

2. [T / F] Two symmetric matrices are orthogonally similar if and only if they have the same characteristic polynomial.

3. [T / F] A negative definite quadratic form has a negative determinant for its associated matrix (for $n \times n$ where n is odd).

4. [T / F] The change of basis matrix P in $x = Py$ must have orthogonal columns of unit length to preserve the geometry of the quadratic form.

5. [T / F] The Principal Axes Theorem cannot be applied to matrices with complex eigenvalues.

PART C: OPEN-ENDED QUESTIONS (30 POINTS)

Answer each question in the space provided. Show your reasoning clearly.

1. **(10 pts)** Describe the step-by-step process of using the Principal Axes Theorem to identify the conic section defined by $4x_1^2 - 4x_1x_2 + x_2^2 = 5$.

2. **(10 pts)** Explain why the symmetry of matrix A is essential for the validity of the Principal Axes Theorem.

3. **(10 pts)** Given a symmetric matrix A, define the criteria for determining if the quadratic form x^TAx is positive definite, negative definite, or indefinite using its eigenvalues.

INSTRUCTOR COPY — ANSWER KEY

Part A — MCQ Answer Key

Q	Answer	Q	Answer
1	**B**	6	**B**
2	**C**	7	**C**
3	**A**	8	**D**
4	**A**	9	**B**
5	**B**	10	**B**

MCQ Explanations:

- **Q1 (B):** For a quadratic form $\sum a_{ij}x_ix_j$, the entries a_{ij} and a_{ji} of the symmetric matrix are each half of the coefficient of x_ix_j. Thus, $4/2 = 2$.
- **Q2 (C):** Orthogonal diagonalization preserves the eigenvalues of the matrix, as P^TAP is similar to A (and specifically orthogonally equivalent).
- **Q3 (A):** A quadratic form is positive definite if and only if all eigenvalues of the associated symmetric matrix are strictly positive.
- **Q4 (A):** The canonical form is $\lambda_1 y_1^2 + \lambda_2 y_2^2 = 1$ when using the Principal Axes Theorem.
- **Q5 (B):** The Principal Axes Theorem guarantees that any symmetric matrix can be orthogonally diagonalized, which is the core of the theorem.
- **Q6 (B):** Sylvester's Law of Inertia states that the number of positive, negative, and zero eigenvalues is invariant under congruence transformations.
- **Q7 (C):** A zero eigenvalue implies the form is degenerate (semi-definite) as it can take the value zero for non-zero vectors.
- **Q8 (D):** Orthogonal diagonalization effectively rotates the coordinate system to align with eigenvectors, which removes cross-terms and yields a sum of squares.
- **Q9 (B):** Since the eigenvalues have opposite signs (3 and -1), the canonical form is $3y_1^2 - y_2^2 = 1$, which represents a hyperbola.
- **Q10 (B):** The theorem allows us to simplify the equation of a quadric surface into a standard form, making it easy to identify the geometry.

Part B — True/False Answer Key

- **TF1 (TRUE):** For any quadratic form x^TAx, one can define a symmetric matrix $B = (A + A^T)/2$ such that $x^TBx = x^TAx$.
- **TF2 (TRUE):** Symmetric matrices are orthogonally diagonalizable; if they share the same eigenvalues (characteristic polynomial), they are orthogonally similar.
- **TF3 (TRUE):** If all eigenvalues are negative, the determinant is the product of n negative values, which is negative for odd n.
- **TF4 (TRUE):** To perform an orthogonal diagonalization, P must be an orthogonal matrix, meaning its columns are orthonormal.

- **TF5 (TRUE):** The spectral theorem for symmetric matrices guarantees all eigenvalues are real; thus, complex eigenvalues do not occur for symmetric matrices.

Part C — Open-Ended Answer Guidance

Open-Ended Q1 (10 pts): Describe the step-by-step process of using the Principal Axes Theorem to identify the conic section defined by $4x_1^2 - 4x_1x_2 + x_2^2 = 5$.

Answer Guidance: 1. Express the quadratic form as x^TAx. 2. Find the eigenvalues of A by solving $det(A - \lambda I) = 0$. 3. Find the corresponding orthonormal eigenvectors. 4. State the canonical form $\lambda_1 y_1^2 + \lambda_2 y_2^2 = 5$. 5. Identify the conic section (e.g., parabola) based on the eigenvalues.

Open-Ended Q2 (10 pts): Explain why the symmetry of matrix A is essential for the validity of the Principal Axes Theorem.

Answer Guidance: 1. Mention the Spectral Theorem for symmetric matrices. 2. Explain that only symmetric matrices are guaranteed to be diagonalizable by an orthogonal matrix. 3. Discuss the importance of having real eigenvalues. 4. Explain how symmetry ensures eigenvectors corresponding to distinct eigenvalues are orthogonal.

Open-Ended Q3 (10 pts): Given a symmetric matrix A, define the criteria for determining if the quadratic form x^TAx is positive definite, negative definite, or indefinite using its eigenvalues.

Answer Guidance: 1. Positive definite: All eigenvalues > 0. 2. Negative definite: All eigenvalues < 0. 3. Indefinite: At least one positive and at least one negative eigenvalue. 4. Note the exclusion of zero eigenvalues for definite cases.